Planetary Gearheads

High Torque



Series 22GPT

Values at 22°C							
Number of gear stages		1	2	2	3	4	4
Reduction ratio [1] (rounded)		3:1 3,6:1 4,5:1 6,6:1	9:1	11:1 14:1 16:1 20:1 24:1 30:1 44:1	41:1 49:1 59:1 72:1 89:1 108:1 131:1 158:1 196:1	178:1 215:1 267:1 323:1 401:1 474:1 588:1 862:1	711:1 1 042:1 1 294:1
Continuous torque, max.	Nm	0,45	0,8	0,8	1,3	1,8	1,4
Intermittent torque, max.	Nm	0,6	1,1	1,1	1,8	2,5	2
Peak torque	Nm	1	2,5	2,5	3,5	4,5	4
Continuous input speed, max.	min-1	9 000	10 000	12 000	15 000	15 000	15 000
Intermittent input speed, max.	min-1	11 000	12 000	15 000	20 000	20 000	20 000
Continuous output power, max.	W	21	12	12	8	7	7
Intermittent output power, max.	W	30	18	18	12	10	10
Efficiency, max.	%	92	84	82	78	65	65
Input inertia with pinion, max.	gmm²	75	78	50	34	14	13
Torsional stiffness, typical	Nm/°	6	11	11	11	11	11
Backlash, at no-load, typical	°	0,8	0,8	0,8	0,8	0,8	0,8
Shaft load, max.:							
– radial (10 mm from mounting face)	N	65	90	90	120	150	150
– axial	N	60	85	85	110	140	140
Shaft press fit force, max.	N	150	150	150	150	150	150
Shaft play:							
– radial (10 mm from mounting face)	mm	≤ 0,05	≤ 0,05	≤ 0,05	≤ 0,05	≤ 0,05	≤ 0,05
– axial	mm	= 0	= 0	= 0	= 0	= 0	= 0
Length without motor L2	mm	19,8	26,2	26,2	32,5	38,9	38,9
Mass without motor and flange	g	44	58	58	72	86	86
Operating temperature range	°C	-30 … +120					
Direction of rotation, drive to output		=					
Housing material		stainless steel					
Geartrain material		stainless steel					
Bearings on output shaft		ball bearings, preloaded					

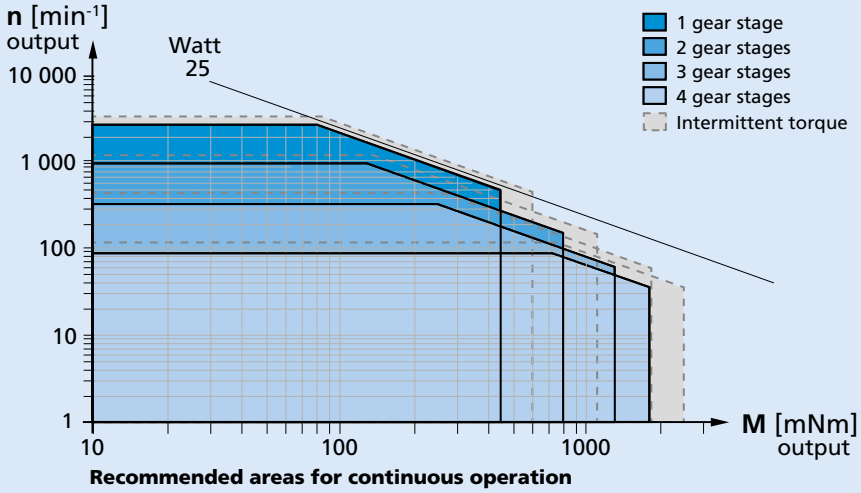
[1] The reduction ratios are rounded, the exact values are available on request or at www.faulhaber.com.

Note: These gearheads are available only with motors mounted.

Note:

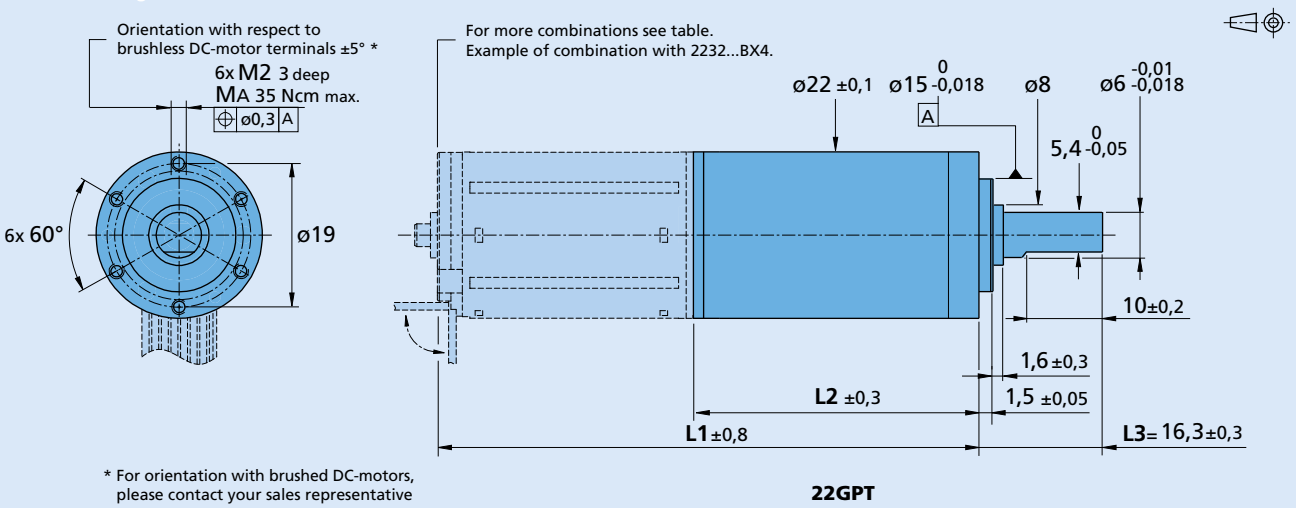
The display shows the range of possible operation points of the drives at a given ambient temperature of 22°C.

The diagram indicates the recommended output speed in relation to the available torque at the output shaft.



© DR. FRITZ FAULHABER GMBH & CO. KG
Specifications subject to change without notice.

Dimensional drawing



* For orientation with brushed DC-motors, please contact your sales representative

22GPT

Options

Example product designation: **22GPT 89:1 KS6KL1**

Option	Type	Description
KS1	Output shaft	Round plain shaft, **L3=** 16,3 mm
KS2	Output shaft	Longer round plain shaft, **L3=** 27 mm
KS3	Output shaft	Shaft with double flat shape of 12 mm length on opposite sides, **L3=** 21 mm
KS4	Output shaft	Shaft with DIN 6885-A keyway and mating key with dimensions 2x2x12 mm, **L3=** 21 mm
KS5	Output shaft	Shaft with 12 mm single flat shape, **L3=** 21 mm
KS6	Output shaft	Shaft with 12 mm single flat shape and 2 mm cross bore at 6 mm of shaft end, **L3=** 21 mm
KS7	Output shaft	Shaft with 10 mm single flat shape and M2.5 axial threaded hole, **L3=** 16,3 mm
KS8	Output shaft	Shaft with fork shape of 2 mm width opening, **L3=** 16,3 mm
KS9	Output shaft	Shaft with KS4 option and M2.5 axial threaded hole, **L3=** 21 mm
KR1	Output shaft/flange	Shaft diameter 4 mm with flat shape 8 mm length and 3x M3 threads on flange, **L3=** 13 mm, **L1/L2+** 0,4 mm (compatible with 22F)
KP1	Ingress Protection	Gearhead with IP54 ingress protection rating (to be combined with specific protected motor)
KL1	Ambient conditions	Low temperature range of -55°C ... +100°C
KL2	Ambient conditions	Vacuum down to 10^{-5} Pa @ 22°C
KL3	Ambient conditions	Temperature range of -55°C ... +150°C and vacuum down to 10^{-9} Pa @ 60°C
KC1	Cable orientation	Motor cable/wires or terminals oriented at 15° CCW vs gearhead front threads
KC2	Cable orientation	Motor cable/wires or terminals oriented at 30° CCW vs gearhead front threads
KC3	Cable orientation	Motor cable/wires or terminals oriented at 45° CCW vs gearhead front threads

Note: Specified values may differ from the standard values depending on the option. Please consult your sales representative for further information.

Product combination

Number of gear stages		1	2	3	4
L2 [mm] = length without motor		19,8	26,2	32,5	38,9
L1 [mm] = length with motor	2224U...SR	45,1	51,5	57,8	64,2
	2232U...SR	53,1	59,5	65,8	72,2
	2237U...CXR	57,9	64,3	70,6	77,0
	2342X...CR	61,8	68,2	74,5	80,9
	2642X...CR/CXR	65,2	71,6	77,9	84,3
	2657X...CR/CXR	80,2	86,6	92,9	99,3
	2668X...CR	91,2	97,6	103,9	110,3
	2232X...BX4	53,6	60,0	66,3	72,7
	2250X...BX4	71,6	78,0	84,3	90,7
	2264X...BP4	87,2	93,6	99,9	106,3
	2214X...BXTH	34,6	41,0	47,3	53,7
	2214X...BXTR	33,8	40,2	46,5	52,9
	3216X...BXTH	36,6	43,0	49,3	55,7
	3216X...BXTR	35,8	42,2	48,5	54,9
	2036U...B	56,9	63,3	69,6	76,0
	2057X...B	76,8	83,2	89,5	95,9
	2444X...B	63,8	70,2	76,5	82,9
	AM2224...10	48,6	55,0	61,3	67,7

© DR. FRITZ FAULHABER GMBH & CO. KG
Specifications subject to change without notice.